Resolution Geometry: General Relativity Without Spacetime

Every GR Prediction Descends to Local Data; Spacetime Is a Representative

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Abstract

We lock the claim that general relativity does not need spacetime as an independent object. Every GR prediction factors through three things:

- the quotient of local data (constraint-satisfying initial data, or link data of a mesh of local systems);
- a selected evolution law;
- a relational readout.

A spacetime manifold with points and coordinates is one representative in a gauge fibre. It is unique up to isometry, and every choice within the fibre gives the same predictions.

The theorem has four parts, each with an executable witness.

(1) *Generation.* Initial data determine their maximal globally hyperbolic development, which is unique up to isometry (Choquet-Bruhat–Geroch). Spacetime is generated, not posited.

(2) *Descent.*

- In FLRW, three lapse gauges give coordinate histories that differ by up to 0.94 at coordinate time 1, while the relational prediction $a(\tau)$ agrees with the exact solution to 10^{-7}.
- In Schwarzschild infall, three charts give coordinate arrival times 40.94, 34.23 and 29.78, while the proper time between the invariant events is 32.409886 in every chart.

(3) *Typing.* Per wavevector there are 6 metric components and 6 momenta. Four constraints and four gauge directions leave 2 physical TT modes (projector rank 2). Points and coordinates are null; the TT data are hidden values.

(4) *Replacement.* The representative can be replaced by a mesh of local systems whose gluing defects are the curvature (Mesh paper). The dynamics can be placed by prediction with reinforcement rather than assumed.

What GR still needs is a selected field equation; the print's no-go stands. The result itself is classical (Einstein's point coincidences, the hole argument, the Cauchy problem, relational observables). RG's contribution is the typed descent statement, the null/hidden split, and the executed witnesses.

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1 The claim, stated so that it can be proved

GR does not need spacetime cannot mean "GR runs without a manifold". Every computation uses some representative. The claim that can be proved is:

Definition 1.1 (Needing an object)**.** A theory *needs* an object X if some prediction of the theory changes when X is replaced by another admissible representative. Otherwise X is a gauge representative, i.e. a point of a fibre over which every prediction is constant.

The paper proves that GR's predictions are constant on the fibre of spacetime representatives. Spacetime is therefore a representative, not a needed object.

2 Generation: spacetime is a development of local data

Imported Theorem 2.1 (Choquet-Bruhat–Geroch [1])**.** *Let (Σ, h_{ij}, K_{ij}) be smooth initial data satisfying the Einstein constraints (with declared matter). They admit a maximal globally hyperbolic development, which is unique up to isometry.*

The input is data on local regions of a slice, i.e. local systems. The output is the spacetime, up to isometry. Whatever is not fixed by the data and the law is diffeomorphism: a gauge, a null direction in RG's sense.

3 Descent: predictions are constant on the representative fibre

Theorem 3.1 (Descent)**.** *Let $\mathcal{O}$ be a relational observable, i.e. a diffeomorphism-invariant function of the development, such as the proper time between invariantly defined events, or a curvature scalar read at the event where a declared clock shows a declared value. Then $\mathcal{O}$ factors through the data modulo spatial diffeomorphisms.*

A quantity that is not relational, such as a metric component at a coordinate point, is not a prediction: it varies along the fibre.

Proof. Combine the Choquet-Bruhat–Geroch uniqueness up to isometry with the invariance of $\mathcal{O}$. A coordinate component changes under a diffeomorphism supported away from the data; this is the hole argument. □

Witness S1 (FLRW dust). Evolve $(\dot{a}/Na)^2 = C/a^3$ in three gauges: $N = 1$, $N = a$, and $N = 1 + \frac{1}{2}\sin 3t$. At coordinate time $t = 1$ the coordinate histories give $a = 1.329$, 0.666 and 1.603. The relational prediction, a as a function of the comoving clock reading τ, matches the exact solution $\left(\frac{3}{2}\sqrt{C}(\tau + \tau_0)\right)^{2/3}$ to 9.9×10^{-8} in all three gauges.

Witness S2 (Schwarzschild infall). Take radial free fall from rest at $r_0 = 10m$ to the event $r = 3m$, which is invariantly defined by the areal radius. Computed by quadrature in three charts:

Chart	Proper time	Coordinate time at arrival
Schwarzschild	32.409886	40.938
Painlevé–Gullstrand	32.409886	34.233
ingoing Eddington–Finkelstein	32.409886	29.779
exact (cycloid)	32.409886	—

The curvature read at arrival, $K = 48m^2/r^6 = 0.0658$, is the same in every chart.

Witness S3 (canary). The coordinate component g_{tt} at $t = 1$ takes different values in the three FLRW gauges. It is a representative component, not a GR prediction.

4 Typing: null versus hidden

Proposition 4.1 (Degrees of freedom; S4)**.** *At linear order the ADM phase space has* $6 + 6$ *components per wavevector. Four constraints and four gauge directions (lapse and shift) remove* $4+4$*, leaving* 2×2*: two transverse-traceless (TT) modes. The TT projector on symmetric* 3×3 *tensors has rank 2 (computed).*

In RG's terms:

- **Null:** spacetime points, coordinates, lapse and shift. They are never observable, and every prediction is constant along them.
- **Hidden:** the two TT modes. They are invisible to sources but real, visible in tidal histories (Both Sides).

GR needs the hidden values. It does not need the null ones, and spacetime as an object consists of exactly the null directions together with a representative.

5 Replacement: meshes of local systems and placed dynamics

- **Mesh paper.** A mesh of locally flat systems with protocol links reconstructs the geometry with no background spacetime: curvature is the gluing defect.
  - On a Schwarzschild slice, spatial curvature from rulers is recovered with median error $1.07\% \to 0.066\%$.
  - The tidal field from clock ratios is recovered to 10^{-6}.
- **Placement battery (uploaded).** A transition law is placed by prediction, with a declared lock: full rank, conditioning, prediction gain over wrong and kinematic models, and contraction of the hidden coupling. Results: 8 of 16 regimes lock. At noise 0.01, the history lengths 2.96 and 4.24 lock (gain 7.2 and 3.8, coupling 0.189 and 0.176, true 0.18). Short histories are rejected, and noise 0.03 is refused at gain 1.49. Placement is not realism.

6 Main theorem

Main Theorem (GR without spacetime). *GR needs exactly three things:*

(i) local data: constraint-satisfying initial data, or mesh link data;

(ii) a selected evolution law: the Einstein equations, or their discrete Regge form;

(iii) relational readouts.

Every GR prediction factors through these. The spacetime manifold with its points and coordinates is a representative of the development, unique up to isometry. Predictions are constant on its fibre, so GR does not need spacetime as an independent object. The physical content consists of the relational observables and the two TT hidden values per point. Points, coordinates, lapse and shift are null.

No-Go 6.1 (What is not claimed).

- The field equation is not derived; it is selected (Lovelock, Regge, H-min). The print's source-remainder no-go stands.
- No ontological claim is made: RG is silent on whether spacetime "exists". The claim is only that GR does not need it.
- The result is classical in substance (Einstein, the hole argument, Choquet-Bruhat–Geroch, relational observables [2, 1, 3]). The novelty is the typed descent and the batteries.
- Not covered: non-globally-hyperbolic developments, quantum gravity, and full Lorentzian mesh dynamics.

7 Battery

- `rg_gr_without_spacetime.py`: S1–S4.
- Imported:
  - the mesh battery (M1–M6);
  - the uploaded placement battery (CSV results audited: 8 of 16 locks reproduced from the lock table).

References

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